The genus-1 seven faces: elliptic collision residues and the KZB connection

Raeez Lorgat

Abstract

At genus 0, the collision residue of a chirally Koszul chiral algebra admits seven equivalent presentations: as a twisting morphism, a line-operator r-matrix, a λ-bracket, commuting sphere Hamiltonians, the Drinfeld r-matrix, the Sklyanin bracket, and the Gaudin Hamiltonians. We prove the genus-1 extension: on an elliptic curve E_τ, the genus-1 collision residue $r^{(1)}_{\mathcal{A}}(z,\tau)$ is the elliptic regularization of the genus-0 residue, obtained by replacing rational propagator kernels with prime-form elliptic kernels. All seven faces at genus 1 specialize a *single E_1-datum* $\Theta_{\mathcal{A}}$ via the dictionary $\Psi \rightsquigarrow \hbar \rightsquigarrow \tau$: Ψ is the algebraic spectral parameter on genus-0 faces, $\hbar$ is the quantisation parameter on the Drinfeld / Sklyanin / Yangian faces, and τ is the elliptic modulus on the KZB / Belavin–Drinfeld / Felder faces. The passage is the bar-intrinsic projection $\Theta^{(g=1)}_{\mathcal{A}} = \mathrm{Reg}_\tau \circ \Theta^{(g=0)}_{\mathcal{A}}$ where Reg_τ is elliptic regularisation. For affine Kac–Moody algebras $\mathcal{A} = \widehat{\mathfrak{g}}_k$, the seven faces specialize to the Knizhnik–Zamolodchikov–Bernard connection, the Belavin–Drinfeld elliptic r-matrix, the Felder dynamical elliptic $R(z,\tau)$ (via Fay trisecant), and the elliptic Gaudin Hamiltonians. The prime-form B-cycle shift is the local source of the Drinfeld–Kohno quantum-group parameter; with $\hbar = (k+h^\vee)^{-1}$, the standard parameter is $q_{\mathrm{DK}} = \exp(\pi i \hbar)$ and the full-cycle factor is q^2_{DK}. For class M algebras (Virasoro, $\mathcal{W}_N$), the collision residue produces higher-order elliptic differential operators involving $P_\tau, P'_\tau, \dots, P^{(2N-3)}_\tau$ outside the affine Hitchin–KZB r-matrix lane.

Remark 0.1 (Genus-1 seven-face datum). The genus-1 comparison kernels are obtained by restricting the bar-intrinsic Maurer–Cartan element to the elliptic collision stratum. The KZB, Belavin–Drinfeld, Felder, and Gaudin readouts are different projections of this restricted datum. The prime-form normalization uses

$$\xi_\tau(z) = \partial_z \log \theta_1(z \mid \tau) = \zeta_\tau(z) - 2\eta_1 z, \qquad P_\tau(z) = -\partial_z \xi_\tau(z) = \wp(z,\tau) + 2\eta_1.$$

The quasi-period $\xi_\tau(z+\tau) - \xi_\tau(z) = -2\pi i$ is the local monodromy input; the quantum-group parameter also requires the chosen Drinfeld–Kohno or Kazhdan–Lusztig normalization.

1 Introduction

The genus-0 seven-face theorem [8] identifies the collision residue $r_{\mathcal{A}}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ of a chirally Koszul chiral algebra $\mathcal{A}$ with seven classical objects: a bar-cobar twisting morphism, a line-operator r-matrix in the Dimofte–Niu–Py framework [4], a classical λ-bracket in the Khan–Zeng 3d sigma-model [6], the Gaiotto–Zeng commuting sphere Hamiltonians, the Drinfeld r-matrix, the Sklyanin Poisson bracket, and the Gaudin Hamiltonians. The seven presentations agree as elements of $\mathrm{End}_{\mathcal{A}}(2)[[z^{-1}]]$; their mutual identification is controlled by the collision depth $k_{\max}$ and the shadow class $G/L/C/M$.

The collision residue $r_{\mathcal{A}}(z)$ is the degree-2 component of the E_1 Maurer–Cartan element $\Theta^{E_1}_{\mathcal{A}}$ in the ordered convolution algebra $\mathfrak{g}^{E_1}_{\mathcal{A}}$, where $B^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the cofree tensor coalgebra with deconcatenation coproduct. Its Σ_n-coinvariant projection is the scalar shadow κ: $\mathrm{av}(r(z))$ in abelian and scalar families, and $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody; the spectral parameter that κ discards is precisely the data the seven faces read. At genus 1, the spectral parameter acquires modular dependence: $r^{(1)}_{\mathcal{A}}(z,\tau)$ depends on the elliptic modulus τ, and the B-cycle monodromy of the ordered collision residue encodes the quantum group parameter.

This paper extends the identification to genus 1. The elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z}+\mathbb{Z}\tau)$ replaces the Riemann sphere, the rational propagator $1/z$ is replaced by the prime-form kernel $\xi_\tau(z) = \theta_1'(z,\tau)/\theta_1(z,\tau)$, and each rational function $1/z^{n+1}$ in the collision expansion is replaced by the corresponding derivative of $P_\tau(z) = -\partial_z \xi_\tau(z)$. The resulting genus-1 collision residue $r^{(1)}_{\mathcal{A}}(z,\tau)$ admits seven equivalent presentations that degenerate to the genus-0 faces as $\tau \to i\infty$.

The main results are:

(1) The *elliptic regularization theorem* (Theorem 2.4): the genus-1 collision residue is

$$r^{(1)}_{\mathcal{A}}(z,\tau) \;=\; c_0\, \xi_\tau(z) \;+\; \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!}\, c_n \cdot P^{(n-1)}_\tau(z),$$

where $c_n = a_{(n)}b$ are the OPE-mode collision coefficients.

(2) The identification of the dz-component of the *KZB connection* with the genus-1 collision residue, and of the $d\tau$-component with its z-derivative (Theorem 3.1).

(3) The identification of the genus-1 collision residue with the *Belavin–Drinfeld elliptic r-matrix* (Theorem 4.1).

(4) The *elliptic Gaudin comparison*: the two-point and Cartan sectors match the KZB kernel; full $n \geq 3$ flatness is the Felder dynamical Yang–Baxter statement (Theorem 5.1).

(5) The *class M collision residue* for Virasoro and $\mathcal{W}_N$: higher-order elliptic operators outside the affine Hitchin–KZB r-matrix lane (Theorems 6.1 and 6.3).

(6) The *degeneration theorem*: all seven genus-1 faces degenerate continuously to their genus-0 counterparts as $q = e^{2\pi i\tau} \to 0$ (Theorem 7.1).

The prime-form kernel defines the elliptic collision residue. Its z- and τ-derivatives give the KZB connection, its root-channel continuation gives the Belavin–Drinfeld kernel, and its collision-depth projections give the Gaudin and class-M operators.

2 The elliptic collision residue

2.1 The genus-1 bar propagator

At genus 0 the bar propagator is $\eta_{12} = d\log(z_1 - z_2)$: a meromorphic one-form on $\mathbb{P}^1$ with a simple pole along the diagonal. On the elliptic curve E_τ, Liouville's theorem forbids a meromorphic function with a single simple pole that is doubly periodic. The propagator must be replaced by a quasi-periodic object.

Definition 2.1 (Genus-1 bar propagator)**.** The *genus-1 bar propagator* on E_τ is the logarithmic derivative of the prime form:

$$\eta^{E_\tau}_{12} \;=\; d\log E(z_1,z_2) \;=\; \xi_\tau(z_1-z_2)\, d(z_1-z_2),$$

where $E(z_1, z_2) = \theta_1(z_1 - z_2 \mid \tau)/\theta_1'(0 \mid \tau)$ is the prime form on E_τ (a section of $K^{-1/2} \boxtimes K^{-1/2}$), θ_1 is the Jacobi theta function, and

$$\xi_\tau(z) := \partial_z \log \theta_1(z \mid \tau) = \frac{\theta_1'(z \mid \tau)}{\theta_1(z \mid \tau)} = \zeta_\tau(z) - 2\eta_1 z$$

is the prime-form logarithmic kernel. Here ζ_τ is the Weierstrass zeta function for periods $(1, \tau)$:

$$\zeta_\tau(z) \;=\; \frac{1}{z} + \sum_{(m,n)\neq(0,0)} \left[\frac{1}{z - m - n\tau} + \frac{1}{m + n\tau} + \frac{z}{(m + n\tau)^2}\right].$$

Put $P_\tau(z) := -\partial_z \xi_\tau(z) = \wp(z, \tau) + 2\eta_1$. The prime-form kernel has Laurent expansion $\xi_\tau(z) = 1/z + O(z)$ near $z = 0$, matching the genus-0 propagator $1/z$ at leading order. The difference is global: ξ_τ is quasi-periodic,

$$\xi_\tau(z + 1) = \xi_\tau(z), \qquad \xi_\tau(z + \tau) = \xi_\tau(z) - 2\pi i.$$

The quasi-periodicity of the propagator is the geometric origin of the B-cycle monodromy that distinguishes genus 1 from genus 0 in all seven faces.

Remark 2.2 (Propagator weight). The bar propagator $d \log E(z, w)$ has weight 1 in both variables, regardless of the conformal weight of the fields being sewed. The logarithmic derivative of the prime form is a $(1, 0)$-form in each variable; the conformal weight h of the generator enters the OPE-mode coefficient c_n, not the propagator kernel.

2.2 The collision residue

Definition 2.3 (Genus-1 collision residue). For a chirally Koszul chiral algebra $\mathcal{A}$ on E_τ, the *genus-1 collision residue* is

$$r^{(1)}_{\mathcal{A}}(z, \tau) \;:=\; \mathrm{Res}^{\mathrm{coll}}_{1,2}(\Theta_{\mathcal{A}})\big|_{E_\tau} \;\in\; \mathrm{End}_{\mathcal{A}}(2)[[z^{-1}]] \otimes \mathbb{C}[[\tau]],$$

where $\Theta_{\mathcal{A}}$ is the bar-intrinsic Maurer–Cartan element.

Theorem 2.4 (Elliptic regularization). *Let $\mathcal{A}$ be a chirally Koszul chiral algebra with genus-0 collision residue $r_{\mathcal{A}}(z) = \sum_{n\geq 0} c_n/z^{n+1}$, where $c_n = a_{(n)}b$ are the OPE-mode coefficients. The genus-1 collision residue is*

$$r^{(1)}_{\mathcal{A}}(z, \tau) \;=\; c_0\, \xi_\tau(z) \;+\; \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!}\, c_n \cdot P^{(n-1)}_\tau(z), \tag{2.1}$$

where $P^{(m)}_\tau = d^m P_\tau/dz^m$. Since $P_\tau = \wp + 2\eta_1$, one has $P^{(m)}_\tau = \wp^{(m)}$ for $m \geq 1$. In particular, the first three terms are

$$r^{(1)}_{\mathcal{A}}(z, \tau) \;=\; c_0\, \xi_\tau(z) \;+\; c_1\, P_\tau(z) \;-\; \tfrac{1}{2}\, c_2\, P'_\tau(z) \;+\; \cdots$$

The Laurent expansion of $r^{(1)}_{\mathcal{A}}$ near $z = 0$ agrees with $r_{\mathcal{A}}(z)$ at leading order.

Proof. Each rational function $1/z^{n+1}$ $(n \geq 0)$ in the genus-0 collision expansion is the unique meromorphic function on $\mathbb{P}^1$ with a pole of order $n + 1$ at $z = 0$ and vanishing at $z = \infty$. On E_τ, the prime-form normalization fixes the corresponding polar kernel. For $n = 0$ it is

$\xi_\tau(z)$; for $n \geq 1$ it is $(-1)^{n-1} P_\tau^{(n-1)}(z)/n!$. The prefactor is fixed by the Laurent expansion $P_\tau^{(m)}(z) = (-1)^m (m+1)!/z^{m+2} + O(1)$ at $m = n-1$:

$$\frac{(-1)^{n-1}}{n!} P_\tau^{(n-1)}(z) \;=\; \frac{(-1)^{n-1}}{n!} \cdot \frac{(-1)^{n-1}\, n!}{z^{n+1}} + O(z^{-n+1}) \;=\; \frac{1}{z^{n+1}} + O(z^{-n+1}).$$

Substituting into the propagator-extraction formula of Definition 2.1 yields (2.1). □

Remark 2.5 (Quasi-periodicity dichotomy). The kernel P_τ and all its derivatives are doubly periodic. The quasi-periodicity of $r^{(1)}_{\mathcal{A}}(z,\tau)$ is therefore inherited entirely from the ξ_τ-term:

$$r^{(1)}_{\mathcal{A}}(z+\tau,\tau) - r^{(1)}_{\mathcal{A}}(z,\tau) \;=\; -2\pi i \cdot c_0. \tag{2.2}$$

When $c_0 = 0$ (the Cartan-type case: $\beta\gamma$, lattice VOAs), the collision residue is a genuine meromorphic function on E_τ. When $c_0 \neq 0$ (the Yangian-type case: affine Kac–Moody, Virasoro, $\mathcal{W}_N$), it is quasi-periodic with non-trivial B-cycle monodromy.

3 Face 4: the KZB connection

At genus 0, Face 4 identifies the collision residue with the Gaiotto–Zeng commuting sphere Hamiltonians. At genus 1, the sphere Hamiltonians are replaced by the Knizhnik–Zamolodchikov–Bernard connection on the configuration space $\mathrm{Conf}_n(E_\tau)$. The KZB connection has two components: a dz-component generalizing the genus-0 KZ connection, and a $d\tau$-component encoding the dependence of conformal blocks on the modulus τ.

The formulas in this section use the KZ-normalized conformal-block kernel. In trace-form normalization the binary affine collision residue is $k\Omega_{\mathrm{tr}}\xi_\tau(z)$. The KZ coefficient is $\Omega_{\mathrm{KZ}}\xi_\tau(z)/(k+h^\vee)$. These are related by the Sugawara normalization; they are not identical functions of the level.

Theorem 3.1 (Face 4: KZB connection). *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at level $k \neq -h^\vee$, the modular shadow connection on $\mathrm{Conf}_n(E_\tau)$ restricts to the KZB connection:*

$$\nabla^{\mathrm{KZB}} \;=\; d \;-\; \underbrace{\frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,\xi_\tau(z_{ij})\,dz_{ij}}_{dz\text{-component: genus-1 collision residue}} \;-\; \underbrace{\frac{1}{k+h^\vee}\sum_{i<j}\Omega_{ij}\,P_\tau(z_{ij})\,d\tau}_{d\tau\text{-component: modular Hamiltonian}},$$

where $z_{ij} = z_i - z_j$ and Ω_{ij} is the Casimir acting on the i-th and j-th tensor factors.

(i) *The dz-component equals the genus-1 collision residue:*

$$H^{\mathrm{KZB}}_{z,i} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,\xi_\tau(z_{ij}) \;=\; \sum_{j\neq i}\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_{ij},\tau)\big]_{ij}. \tag{3.1}$$

(ii) *The $d\tau$-component equals the z-derivative of the collision residue, through the identity $P_\tau(z) = -\xi'_\tau(z)$:*

$$H^{\mathrm{KZB}}_{\tau,i} \;=\; \frac{1}{k+h^\vee}\sum_{j\neq i}\Omega_{ij}\,P_\tau(z_{ij}) \;=\; -\sum_{j\neq i}\partial_z\big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_{ij},\tau)\big]_{ij}. \tag{3.2}$$

Proof. **Part (i).** The affine Kac–Moody OPE $J^a(z)J^b(w) \sim k\,\delta^{ab}/(z-w)^2 + f^{ab}_{\ \ c}J^c(w)/(z-w)$ has $c_0 = \Omega/(k+h^\vee)$ and $c_n = 0$ for $n \geq 1$: the bar propagator absorbs one pole order, so the only surviving term is c_0. By Theorem 2.4, the elliptic regularization gives $r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) = \Omega\,\xi_\tau(z)/(k+h^\vee)$ in KZ normalization. Substituting into the Gaiotto–Zeng formula produces (3.1), which is the dz-component of the KZB connection [3, 7].

Part (ii). The $d\tau$-component involves the modular Hamiltonian D_i, whose leading term is $\sum_{j\neq i} P_\tau(z_{ij})\,\Omega_{ij}/(k+h^\vee)$. The identity $P_\tau = -\xi'_\tau$ identifies this with $-\partial_z r^{(1)}_{\mathcal{A}}$, yielding (3.2). □

Remark 3.2 (Shadow connection interpretation). The KZB connection is the genus-1 instance of the modular shadow connection $\nabla^{\mathrm{hol}}_{g,n}$ on $\overline{\mathcal{M}}_{g,n}$: at genus 0 it gives the KZ connection, at genus 1 the KZB connection, and at genus ≥ 2 it produces new flat connections on configuration spaces of higher-genus curves.

4 Face 5: the elliptic r-matrix

At genus 0, Face 5 identifies the collision residue with the KZ-normalized rational kernel $\Omega/((k+h^\vee)z)$. At genus 1, the rational r-matrix is replaced by the classical elliptic r-matrix of Belavin [1] and Belavin–Drinfeld [2].

Theorem 4.1 (Face 5: elliptic r-matrix). *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ with $\mathfrak{g}$ simple and $k \neq -h^\vee$, the KZ-normalized genus-1 two-point kernel equals the classical Belavin elliptic kernel up to the standard level normalization:*

$$r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) \;=\; \frac{1}{k+h^\vee}\, r^{\mathrm{ell}}_{\mathfrak{g}}(z,\tau). \tag{4.1}$$

For $\mathfrak{g} = \mathfrak{sl}_2$, the Belavin r-matrix is

$$r^{\mathrm{ell}}_{\mathfrak{sl}_2}(z,\tau) \;=\; \xi_\tau(z)\cdot\frac{H\otimes H}{2} \;+\; \phi_+(z,\tau)\,E\otimes F \;+\; \phi_-(z,\tau)\,F\otimes E,$$

where $\{H, E, F\}$ is the standard basis of $\mathfrak{sl}_2$ and the theta-function ratios are

$$\phi_\pm(z,\tau) \;=\; \frac{\theta_1'(0\mid\tau)\,\theta_1(z\pm\frac{1}{2}\mid\tau)}{\theta_1(z\mid\tau)\,\theta_1(\pm\frac{1}{2}\mid\tau)}. \tag{4.2}$$

The full $n \geq 3$ KZB flatness statement uses the Felder dynamical shift and the dynamical Yang–Baxter equation. It is not a consequence of applying the ordinary rational CYBE to the displayed two-point kernel.

Proof. For the Cartan channel: the Kac–Moody OPE with $c_0 = \Omega/(k+h^\vee)$ gives, after genus-1 regularization, $\xi_\tau(z)\cdot H\otimes H/2$ in the Cartan direction.

For the root channels: on E_τ, the chiral fields J^E, J^F associated to the positive and negative roots satisfy twisted boundary conditions under B-cycle monodromy. The collision residue in the root channels acquires the theta-function ratios $\phi_\pm(z,\tau)$ (4.2), which are the unique meromorphic functions on $\mathbb{C}$ with a simple pole at $z=0$ matching $1/z$ and with B-cycle quasi-periodicity dictated by the root lattice [1].

The full n-point flatness statement belongs to the Felder dynamical KZB lane. The two-point identification (4.1) follows from Theorem 2.4 combined with uniqueness of the classical elliptic r-matrix [2]. □

Remark 4.2 (Higher rank). For $\mathfrak{g} = \mathfrak{sl}_N$ with $N \geq 3$, the Belavin elliptic r-matrix is a matrix-valued meromorphic function on E_τ with poles at $z=0$ whose residue is the $\mathfrak{sl}_N$ Casimir. The collision residue produces this r-matrix by the same mechanism: elliptic regularization of the Cartan and root channels, with theta-function ratios $\phi_\alpha(z,\tau)$ indexed by the roots α. The identification (4.1) holds for all simple $\mathfrak{g}$.

5 Face 7: the elliptic Gaudin model

At genus 0, Face 7 identifies the collision residue with the Gaudin Hamiltonians $H_i^{\mathrm{Gaudin}} = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$. At genus 1, the Gaudin model acquires elliptic structure.

Theorem 5.1 (Face 7: elliptic Gaudin kernel). *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$, the KZ-normalized two-point elliptic Gaudin kernel on* $\mathrm{Conf}_n(E_\tau)$ *is*

$$H_i^{\mathrm{ell}} \;=\; \frac{1}{k+h^\vee} \sum_{j\neq i} \Omega_{ij}\, \xi_\tau(z_i - z_j) \;=\; \sum_{j\neq i} \big[r^{(1)}_{\widehat{\mathfrak{g}}_k}(z_i - z_j, \tau) \big]_{ij}.$$

After adding the Felder dynamical shift, this kernel is the two-point sector of the flat KZB/Gaudin connection.

Proof. The identification of H_i^{ell} with the collision residue is immediate from Theorem 2.4: $r^{(1)}_{\widehat{\mathfrak{g}}_k}(z,\tau) = \Omega\, \xi_\tau(z)/(k+h^\vee)$ in KZ normalization. For $n \geq 3$, flatness of the elliptic Gaudin connection requires the dynamical Cartan variable and Felder's dynamical Yang–Baxter equation; the non-dynamical two-point kernel displayed here is the affine Cartan/KZB comparison surface. □

Remark 5.2 (Hitchin system). The elliptic Gaudin model is the genus-1 instance of the Hitchin integrable system on $\mathrm{Bun}_G(E_\tau)$. The Hitchin Hamiltonians are sections of a line bundle on the base of the Hitchin fibration; at genus 1, the base is the affine space of elliptic spectral curves, and the Hitchin Hamiltonians restrict to the elliptic Gaudin Hamiltonians on the trivial G-bundle. At genus $g \geq 2$, the collision residue on Σ_g produces the Hitchin Hamiltonians on $\mathrm{Bun}_G(\Sigma_g)$; this is the structural target of the holographic construction [8].

6 Class M at genus 1: higher-order elliptic operators

For affine Kac–Moody algebras (class L), the collision expansion terminates at c_0: the collision depth is $k_{\max} = 1$, and only the ξ_τ-term contributes. For class M algebras (Virasoro, $\mathcal{W}_N$), the OPE has poles of higher order, and the collision expansion involves P_τ, P'_τ, and higher prime-form derivatives.

6.1 Virasoro at genus 1

The Virasoro algebra has maximal OPE pole order $p_{\max} = 4$ (from the $T(z)\,T(w)$ OPE). The bar propagator absorbs one order, giving collision depth $k_{\max} = 3$. The genus-0 collision residue is

$$r_{\mathrm{Vir}_c}(z) \;=\; \frac{c/2}{z^3} \;+\; \frac{2T}{z},$$

with $c_0 = 2T$, $c_1 = 0$ (the z^{-2} pole is absent from the r-matrix by a parity argument: the $d\log$ extraction sends z^{-2n} to $z^{-(2n-1)}$), and $c_2 = c/2$.

Theorem 6.1 (Virasoro genus-1 collision residue). *The genus-1 collision residue of* Vir_c *is*

$$r^{(1)}_{\mathrm{Vir}_c}(z,\tau) \;=\; 2T\cdot\xi_\tau(z) \;-\; \frac{c}{4}\,P'_\tau(z). \tag{6.1}$$

This is an elliptic differential operator of order 2 on conformal blocks on E_τ: the ξ_τ-term acts by multiplication, while the P'_τ-term acts as a second-order differential operator through the stress-energy Ward identity.

Proof. The genus-0 collision expansion has $c_0 = 2T$, $c_1 = 0$, $c_2 = c/2$. By Theorem 2.4:

$$r^{(1)}_{\mathrm{Vir}_c}(z,\tau) \;=\; c_0\,\xi_\tau(z) + c_1\,P_\tau(z) - \tfrac{1}{2}\,c_2\,P'_\tau(z).$$

Since $c_1 = 0$, the P_τ-term vanishes, and the result is (6.1). The Laurent expansion near $z = 0$ confirms consistency: $2T/z - (c/4)\cdot(-2/z^3 + O(z)) = (c/2)/z^3 + 2T/z + O(1)$, matching $r_{\mathrm{Vir}_c}(z)$ at leading order.

The P'_τ-term acts as a second-order differential operator because the Virasoro OPE at order $(z-w)^{-4}$ involves $(c/2)\cdot\mathrm{id}$, which through the Ward identity produces a second derivative ∂_z^2 on the correlator; the elliptic regularization replaces $1/z^4$ by $-P'_\tau(z)/2$. □

Remark 6.2 (The $c_1 = 0$ vanishing). The vanishing of c_1 is a parity phenomenon: the OPE mode $T_{(2)}T = \partial T$ is a total derivative, and its contribution to the r-matrix vanishes after d log-extraction. In the elliptic regularization, this means the $P_\tau(z)$-term is absent. The first non-trivial correction beyond ξ_τ involves P'_τ, not P_τ.

6.2 $\mathcal{W}_N$ at genus 1

The $\mathcal{W}_N$ algebra has generators $W_2 = T, W_3, \ldots, W_N$ of conformal weights $2, 3, \ldots, N$. The maximal OPE pole order among generating fields is $p_{\max} = 2N$ (from the W_N-W_N OPE), giving collision depth $k_{\max} = 2N - 1$.

Theorem 6.3 ($\mathcal{W}_N$ genus-1 collision residue). *The genus-1 collision residue of $\mathcal{W}_N$ involves the prime-form kernel and the derivatives of P_τ up to $P_\tau^{(2N-3)}$:*

$$r^{(1)}_{\mathcal{W}_N}(z,\tau) \;=\; c_0\,\xi_\tau(z) \;+\; \sum_{\substack{n=1\\ c_n\neq 0}}^{2N-2} \frac{(-1)^{n-1}}{n!}\,c_n\cdot P_\tau^{(n-1)}(z), \tag{6.2}$$

where c_n are the OPE-mode collision coefficients. The highest-order term is an elliptic differential operator of order $2N-2$. For $\mathcal{W}_3$: the collision expansion involves $\xi_\tau, P_\tau, P'_\tau, P''_\tau, P'''_\tau$, producing fourth-order elliptic differential operators.

Proof. Direct specialization of Theorem 2.4 to the $\mathcal{W}_N$ OPE data with $p_{\max} = 2N$: the collision depth $k_{\max} = 2N-1$ means the genus-0 expansion $r_{\mathcal{W}_N}(z) = \sum_{n=0}^{2N-2} c_n/z^{n+1}$ has $2N-1$ terms, each regularized by $1/z \mapsto \xi_\tau(z)$ and $1/z^{n+1} \mapsto (-1)^{n-1}P_\tau^{(n-1)}(z)/n!$ for $n \geq 1$. □

Remark 6.4 (Outside the affine KZB lane). The elliptic differential operators produced by the class M collision residue lie outside the affine Hitchin–KZB r-matrix lane. The Hitchin system on $\mathrm{Bun}_G(E_\tau)$ sees the affine Casimir channel. The KZB connection sees the prime-form kernel and its first polar derivative in the standard affine sector. The class M collision residue at depth ≥ 3 produces operators involving $P'_\tau, P''_\tau, \ldots, P_\tau^{(2N-3)}$ from higher OPE poles specific to $\mathcal{W}$-algebras.

7 Degeneration to genus 0

Theorem 7.1 (Degeneration)**.** *As* $\mathrm{Im}(\tau) \to \infty$ *(equivalently,* $q = e^{2\pi i \tau} \to 0$*), all seven genus-1 faces degenerate to their genus-0 counterparts:*

(i) (Propagator.) $\xi_\tau(z) \to \pi \cot(\pi z) \to 1/z + O(z)$.

(ii) (Prime-form derivatives.) $P_\tau(z) \to \pi^2/\sin^2(\pi z) \to 1/z^2 + O(1)$*, and* $P_\tau^{(m)}(z) \to (-1)^m (m+1)!/z^{m+2} + O(z^{-m})$.

(iii) (KZB $\to$ KZ.) *The dz-component degenerates to the KZ connection:* $\xi_\tau(z_{ij}) \to 1/z_{ij}$*. The $d\tau$-component has no genus-0 counterpart.*

(iv) (Elliptic $\to$ rational r-matrix.) *In trace form,*

$$r_{\mathrm{tr}}^{\mathrm{KM}}(z) = k\,\Omega_{\mathrm{tr}}/z.$$

The KZ-normalized connection uses

$$r^{\mathrm{KZ}}(z) = \Omega_{\mathrm{KZ}}/((k+h^\vee)z), \qquad \Omega_{\mathrm{KZ}} = 2h^\vee \Omega_{\mathrm{tr}}.$$

The elliptic kernel degenerates through the trigonometric kernel to the corresponding rational kernel in the chosen normalization. These are conventionally related normalizations, not an identity as rational functions of k.

(v) (Elliptic $\to$ rational Gaudin.) $H_i^{\mathrm{ell}} \to H_i^{\mathrm{Gaudin}} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$.

Proof. As $q \to 0$, the prime-form kernels have the standard degenerations. The logarithmic derivative satisfies $\xi_\tau(z) = \pi \cot(\pi z) + O(q)$, giving $1/z + O(z)$ for $|z| < 1$. Its derivative gives $P_\tau(z) = \pi^2/\sin^2(\pi z) + O(q)$, hence $1/z^2 + O(1)$. Higher derivatives degenerate analogously.

The KZB degeneration: with $\xi_\tau(z) \to 1/z$, the dz-component (3.1) becomes $\sum_{j\neq i} \Omega_{ij}/((k+h^\vee)z_{ij})\,dz_{ij}$, which is the KZ connection.

The elliptic r-matrix degeneration: as $q \to 0$, $\phi_\pm(z,\tau) \to \pi/\sin(\pi z) + O(q)$, recovering the trigonometric r-matrix. A further limit $z \mapsto z/L$, $L \to \infty$ gives $\pi/\sin(\pi z/L) \to L/z$, recovering the rational r-matrix $\Omega/((k+h^\vee)z)$ in the KZ normalization. Both limits are continuous in the topology of formal Laurent series. □

The degeneration theorem provides the consistency check between genus-0 and genus-1: every genus-1 face is a continuous deformation of the corresponding genus-0 face, with deformation parameter q. At $q = 0$, the elliptic curve degenerates to the nodal rational curve, and all seven identifications collapse to their genus-0 ancestors.

8 Computational verification

The genus-1 seven-face checks are implemented by the compute engine `theorem_genus1_seven_faces_engine.py`, which implements the following tests.

(1) **Elliptic regularization.** For each standard family $\mathcal{A}$ in the landscape census, verify that the genus-1 collision residue (2.1) matches the elliptic regularization of the genus-0 collision expansion at $O(q^0)$ within the stated numerical tolerance.

(2) **KZB identification.** For $\mathcal{A} = \widehat{\mathfrak{sl}}_2$ at levels $k = 1, 2, 3, 5, 10$, verify that the dz-component of the KZB connection equals the collision residue, and that the $d\tau$-component equals $-\partial_z r^{(1)}_{\mathcal{A}}$.

(3) **Dynamical closure.** Check the two-point and Cartan sectors of the affine elliptic kernel. The full $n \geq 3$ flatness statement is the Felder dynamical Yang–Baxter closure and is not claimed by this ordinary-kernel test.

(4) **Gaudin kernel.** Check that the two-point and Cartan elliptic Gaudin kernels agree with the KZB kernel in the chosen normalization. Full $n \geq 3$ commutativity belongs to the dynamical KZB/Gaudin lane.

(5) **Degeneration.** Verify that $\|r^{(1)}_{\mathcal{A}}(z, \tau) - r_{\mathcal{A}}(z)\|/\|r_{\mathcal{A}}(z)\| < 10^{-6}$ for $\mathrm{Im}(\tau) \geq 12$ across all standard families.

(6) **Class M structure.** For Vir_c at $c = 1, 1/2, 25, 26$, verify (6.1) and confirm the $c_1 = 0$ vanishing. For $\mathcal{W}_3$ at $c = 2, 50, 98$, verify (6.2) and confirm terms through P'''_τ.

9 The Verlinde formula from ordered chiral homology

At integer level $k \in \mathbb{Z}_{\geq 1}$, the ordered chiral homology of $\widehat{\mathfrak{g}}_k$ on E_τ recovers the Verlinde partition function. The connection to the genus-1 seven-face identification is direct: the KZB connection of Theorem 3.1 has finite monodromy at integer level, and the trace of the B-cycle monodromy over the space of conformal blocks recovers the Verlinde formula.

Proposition 9.1 (Verlinde from genus-1 collision residue). *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at integer level $k \geq 1$, the B-cycle shift of the KZ-normalized prime-form kernel is*

$$\mathcal{M}_B^{\mathrm{ker}} = \exp\Big(-\frac{2\pi i}{k + h^\vee}\,\Omega\Big) = \exp(-2\pi i\, c_0),$$

where $c_0 = \Omega/(k + h^\vee)$ is the degree-0 collision coefficient. The quantum group parameter is $q_{\mathrm{DK}} = \exp(\pi i/(k + h^\vee))$, with full-cycle factor $q_{\mathrm{DK}}^2 = \exp(2\pi i/(k + h^\vee))$. The Verlinde partition function at genus 1 is

$$Z_1(\widehat{\mathfrak{g}}_k) = \sum_{\lambda \in P_+^k} |S_{0\lambda}|^0 = |P_+^k|,$$

where P_+^k is the set of integrable highest weights at level k and $S_{0\lambda}$ are the modular S-matrix entries. At higher genus g, the Verlinde formula $Z_g = \sum_\lambda S_{0\lambda}^{2-2g}$ is recovered from the ordered chiral homology via handle attachment and separating factorization of the ordered bar complex.

Remark 9.2 (The B-cycle monodromy and the quantum group). The quasi-periodicity (2.2) of the genus-1 collision residue is the geometric origin of the quantum group parameter. For affine $\widehat{\mathfrak{sl}}_{2,k}$: the B-cycle monodromy eigenvalues are written in the Drinfeld–Kohno normalization with $q_{\mathrm{DK}} = e^{\pi i/(k+2)}$, and the full cycle contributes q_{DK}^2. The passage from the E_1 collision residue $r^{(1)}(z, \tau)$ to the quantum group $U_{q_{\mathrm{DK}}}(\mathfrak{g})$ is the B-cycle monodromy of the ordered bar complex: the spectral parameter z (genus-0 data) becomes the quantum group parameter through exponentiation of the quasi-period.

10 The critical level

The genus-1 seven-face identification has a distinguished boundary: the critical level $k = -h^\vee$, where the Sugawara construction degenerates. The collision residue and the KZB connection undergo a qualitative change at this value.

Theorem 10.1 (Critical level jump at genus 1). *For $\mathcal{A} = \widehat{\mathfrak{g}}_k$ at the critical level $k = -h^\vee$:*

(i) *The shadow invariant vanishes: $\kappa(\widehat{\mathfrak{g}}_{-h^\vee}) = 0$. The KZB-normalized kernel is singular at this level, so the non-critical monodromy formula has no regular value there.*

(ii) *The degree-2 ordered chiral homology on $V \otimes V$ doubles:* $\dim H^1_{\mathrm{ord},2}$ *jumps from* 4 *(generic level) to* 8 *(critical level) for* $\mathfrak{g} = \mathfrak{sl}_2$ (*finite computation,* $\chi_{\mathrm{ord},2} = -4$ *conserved*). *This is distinct from the full symmetric bar cohomology at critical level, addressed in (iii).*

(iii) *Under the completed critical bar–Ext comparison, the critical bar–Ext groups identify with differential forms on the space of $\mathfrak{g}^\vee$-opers:*

$$H^n_{\mathrm{bExt}}(\widehat{\mathfrak{g}}_{-h^\vee}) \cong \Omega^n(\mathrm{Op}_{\mathfrak{g}^\vee}(D)).$$

This is a completed critical statement, not an equality for the raw symmetric bar cohomology.

(iv) *The genus-*1 *collision residue degenerates: the KZ normalization $r^{(1)}(z,\tau) = \Omega\,\xi_\tau(z)/(k + h^\vee)$ has a pole at $k = -h^\vee$, and the KZB connection becomes singular. The critical level requires a separate regularization via the Feigin–Frenkel center* [5].

Remark 10.2 (Koszulness at critical level). At generic level $k \neq -h^\vee$, the affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ is chirally Koszul, and the seven-face identification applies without modification. At the critical level $k = -h^\vee$, the KZ/KZB normalization and Sugawara topologisation fail, and the Feigin–Frenkel centre appears. This does not by itself negate PBW Koszulness of the universal affine algebra; it moves the statement to the separate critical-level bar-cohomology lane. The seven KZ faces degenerate: the Gaudin Hamiltonians become the Feigin–Frenkel higher Gaudin Hamiltonians, and the r-matrix limit requires the Langlands dual group ${}^L G$ via the geometric Langlands correspondence. The critical level is a boundary of the non-critical seven-face identification.

Remark 10.3 (Spectral sequence at critical level). The passage from generic to critical level is controlled by a spectral sequence with $d_1 = \lambda \cdot [\delta]$, where $\lambda = k + h^\vee$ is the level shift and δ is the boundary class. At $\lambda = 0$ (critical level), d_1 vanishes, and the spectral sequence degenerates at E_1: the bar cohomology jumps because the differential that would kill the extra classes is absent. This is the mechanism by which the degree-2 ordered chiral homology $H^1_{\mathrm{ord},2}$ doubles from 4 to 8 for $\mathfrak{sl}_2$. For the completed critical bar–Ext cohomology at critical level, the d_1-vanishing lets the entire de Rham algebra $\Omega^*(\mathrm{Op}_{\mathfrak{sl}_2}(D))$ appear in completed critical bar–Ext cohomology, which is infinite-dimensional.

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